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Pip

A lightweight AI assistant window with safe local tool access

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INDONESIA

iTECH – Software / Programming

Research Question

Can a free-tier cloud AI chat service be given controlled, auditable access to a student's local files using only local components, and how much of the risk surface can a user-controlled permission layer close?

Aim

Build a desktop window that reuses the AI account a student already has, and gives that AI access to their own files under a boundary the student controls. Evaluate on three axes: does it function, what does it cost, what does it stop.

Hypothesis / Engineering Goal

H1 Capability: a free account can call a local tool server with no API key – Supported. H2 Boundary: a user-controlled permission layer closes most of the risk surface – Partly. H3 Efficiency: converting a document to Markdown reduces token cost – Rejected, conversion made prompts 12% larger. H3 revised: reading fewer pages reduces cost – Supported.

Background Research

Model Context Protocol is an open standard for exposing tools to an AI client. Tauri renders the system webview with a Rust backend, so spawning a process is an explicit command. MarkItDown, Microsoft's document converter, is used as one tool rather than rebuilt. Existing AI desktop clients set the boundary for the vendor and assume a paid tier.

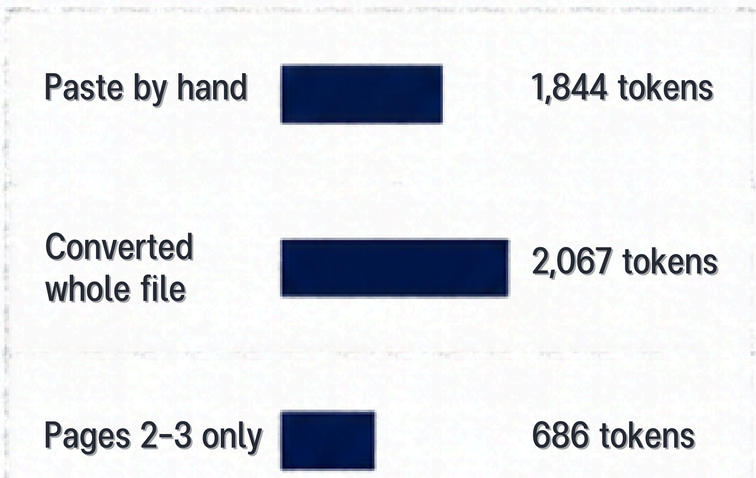
Methods / Procedures

Built in three tiers over seven days so each tier is a complete project if time runs out; the riskiest unknown was tested on day two. Evaluated on three axes: function (the chain split into seven links, a link counts only when observed running), safety (nine adversarial cases, a case passes only if refused and logged), cost (marginal cost to the student, with token efficiency as a sub-metric).

Results

Seven tools, six of seven chain links observed passing, one pending. Eight of nine boundary attacks refused and logged, one open.

Data Graph / Table



One tool measured in isolation: the document converter. Converting alone makes the prompt 12% larger; selecting two pages cuts it 63%.

Discussion of Results

The converter is a utility, not the contribution: MarkItDown behind a thin wrapper, one of seven tools. Conversion makes prompts larger, because structure costs characters. What saves is sending less, and that is only possible because the file stays on the machine.

Errors and Limitation Gaps

No delivery: Pip prepares the prompt, the paste is manual. Tier 3 needs the internet. No user study. macOS only. A folder can be swapped between validation and conversion; that race is open. Not packaged.

Conclusion

A student on a free tier can have agent-like access to their own files without paying for it, with a boundary they set, can revoke, and can read back afterwards.

Future Research

Package the application. Administer the survey with ten to twenty students. OCR for scanned PDFs. Windows support.

References

Model Context Protocol specification; Microsoft MarkItDown; Tauri 2 documentation; MCP Authorization specification.

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